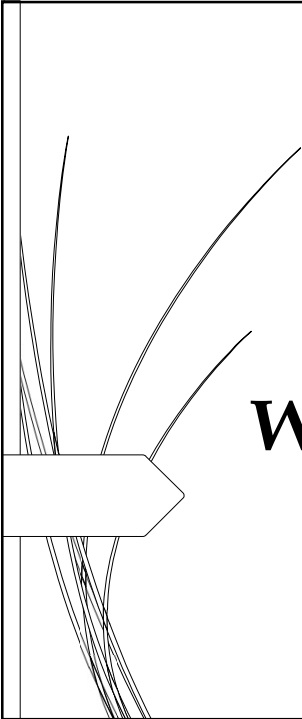




WEB HOSTING PROCESS



What is web hosting?

- Website hosting is a service that allows organizations, companies, and individuals to create and post a website onto the internet
- All websites on the internet need a **web host** to guarantee their availability on the world wide web.
- These websites are hosted, or stored, on unique computer systems known as **servers**.

Steps involved in Hosting

For your website to go live, you must go through the following processes:

1.
Decide on
type of
website
{Static or
Dynamic}

2.
Choose
hosting
server
{Linux or
Windows}

3.
Choose
hosting
plan

4.
Change
your DNS
Address

5.
Upload your
website files

Types of Web Hosting Services

- Web Hosting companies provide special packages, features, and level of management offers to their clients.
- The following are the different types of web hosting services.
 1. Shared hosting
 2. VPS hosting
 3. Dedicated hosting
 4. Cloud hosting

Shared Hosting

- In shared hosting, multiple websites reside on one single physical server and share its resources (CPU, RAM, and Disk space).
- It is very affordable; easy to set up (usually comes with a Control Panel app like cPanel).
- It is best for beginners because it saves cost.
- The shared resources are limited when using shared hosting.
- You could suffer from the "Bad Neighbor Effect"—if one website on the server gets a massive traffic spike or a virus, your site might slow down or crash.

VPS hosting

- In Virtual Private Server(VPS) hosting, a single physical server is divided into multiple "virtual" servers using a technique known as Virtualization.
- Even though you share the hardware, you have your own dedicated slice of resources and your own OS.
- It should be your preferred option if you have high-security concerns but don't want to invest in a faster (but more expensive) dedicated server.
- VPS is best for growing businesses, e-commerce sites, or custom web apps
- It requires more technical knowledge to manage the server software
- VPS hosting is typically more expensive than shared hosting.

Dedicated hosting

- In a dedicated hosting, you rent an entire physical server all to yourself.
- It guarantees maximum performance, total security control, and massive storage.
- Dedicated hosting is the ideal solution for larger businesses and high-traffic websites because it allows for maximum customization, configuration, installation, and flexibility.
- One of the advantages of going this route is that no other website competes with you on all of the resources or pose any sort of security threat.
- Dedicated hosting offers you an entire server for yourself, thereby making it faster, more secure, and more expensive.

Cloud hosting

- Similar to VPS hosting, cloud hosting allows multiple virtual servers (clouds) to work together to host a website or a group of websites.
- Instead of one server, your website is hosted on a network (a "cloud") of many interconnected servers, and should one server fails; another takes over instantly.
- It offers unlimited ability to handle sudden traffic spikes.
- A cloud-hosted website is not limited to a single server, and the resources allocated to it can shrink or expand dynamically, depending on how much traffic you get.
- It's a great option for large websites, including e-commerce websites, newsletters and blogs.
- One advantage of choosing cloud hosting is the possibility for higher scalability.

Uploading files to the cPanel

- cPanel is an online Linux-based web hosting control panel.
- It provides a graphical user interface, and is packed with lots of features and automation tools designed to simplify the process of hosting websites and web applications.

General Steps for uploading Files to cPanel

- To upload a file through cPanel's File Manager, the following are the steps involved
 1. Login to cPanel.
 2. Under Files, click **File Manager**.
 3. Select **Web Root** and **Show Hidden Files**, then click **Go**.
 4. Click to open the folder to which you'd like to upload files.
 5. Click **Upload** from the top toolbar.
 6. Click **Browse** (some web browsers will have **Choose File** instead).
 7. Select the file on your computer that you wish to upload.
 8. Once the file has been selected, it will automatically upload. You may select additional files while uploads are in progress.

Accessing the cPanel -- 1

- **Step 1: Accessing the cPanel Login Page**
- There are three common ways to find your login screen. Replace yourdomain.com with your actual domain name or server IP address you received from your service provider:
- **Method A:** https://yourdomain.com/cpanel
- **Method B:** https://yourdomain.com:2083 (The secure default port)
- **Method C:** Via your Hosting Provider's **Client Area** (usually a "Login to cPanel" button that logs you in automatically).

Account / Products / Manage Hosting

Web Hosting Dashboard

Web Hosting Plan
Monthly Startup Hosting

Expiration Date
29 Nov 2018

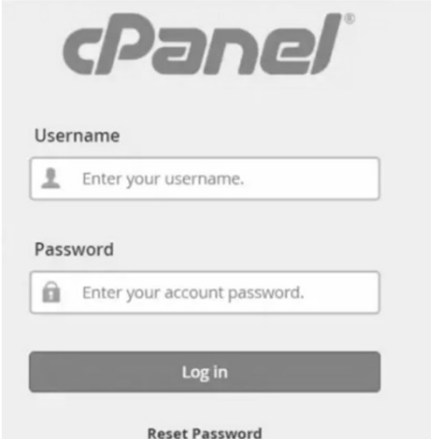
Automatic Renewal
Pending

cPanel Login

Quick Links ▾

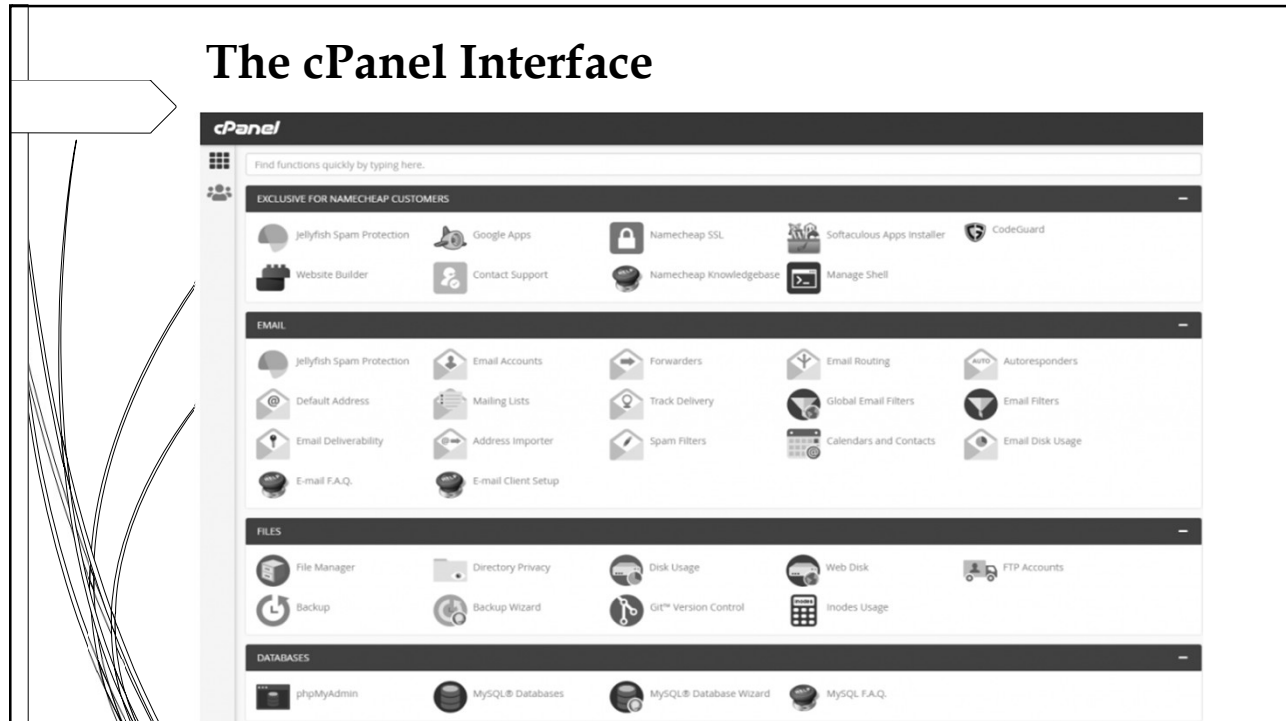
Accessing the cPanel -- 2

- **Step 2: Logging In**
- Enter your **Username** and **Password** (sent to you in your "Welcome/Technical Details" email from your host).
- Click **Log in**.
 - *Note:* If you get an "IP Address Has Changed" error, it's often due to a shifting local network; try logging in through your hosting provider's portal instead.



The image shows the cPanel login interface. At the top is the cPanel logo. Below it are two input fields: 'Username' with a person icon and 'Password' with a lock icon. Below these fields is a 'Log in' button. At the bottom, there is a 'Reset Password' link.

The cPanel Interface



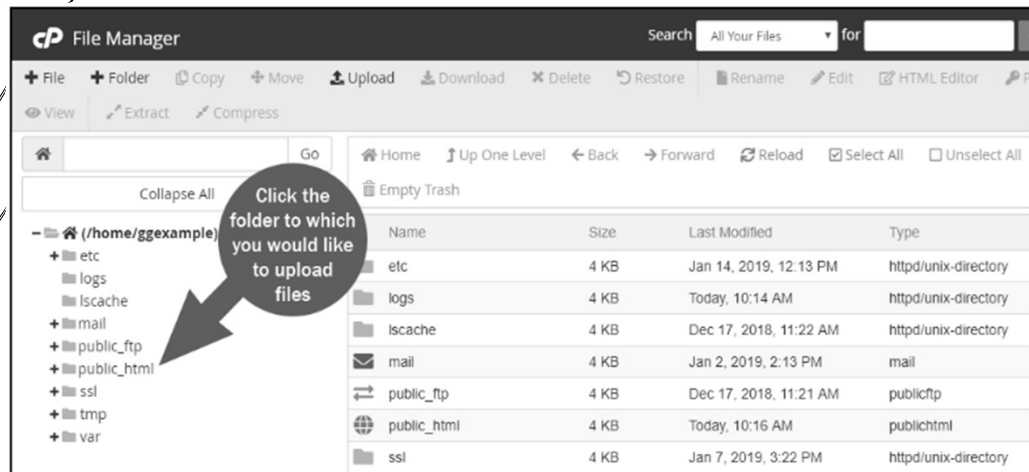
Accessing the cPanel -- 3

- **Step 3: Navigating to the File Manager**
- Once inside the dashboard, locate the **Files** section.
- Click on the **File Manager** icon. *



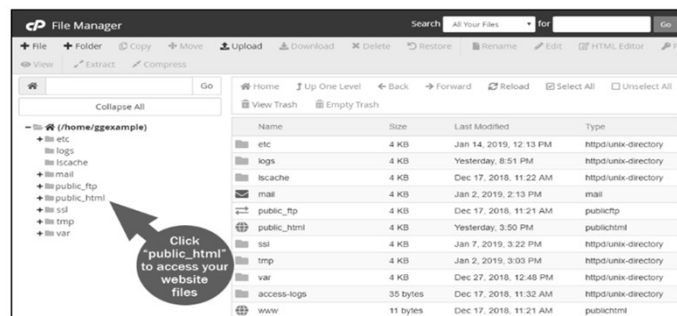
On the left side, click the folder to which you would like to upload files.

The folder contents will be displayed on the right side

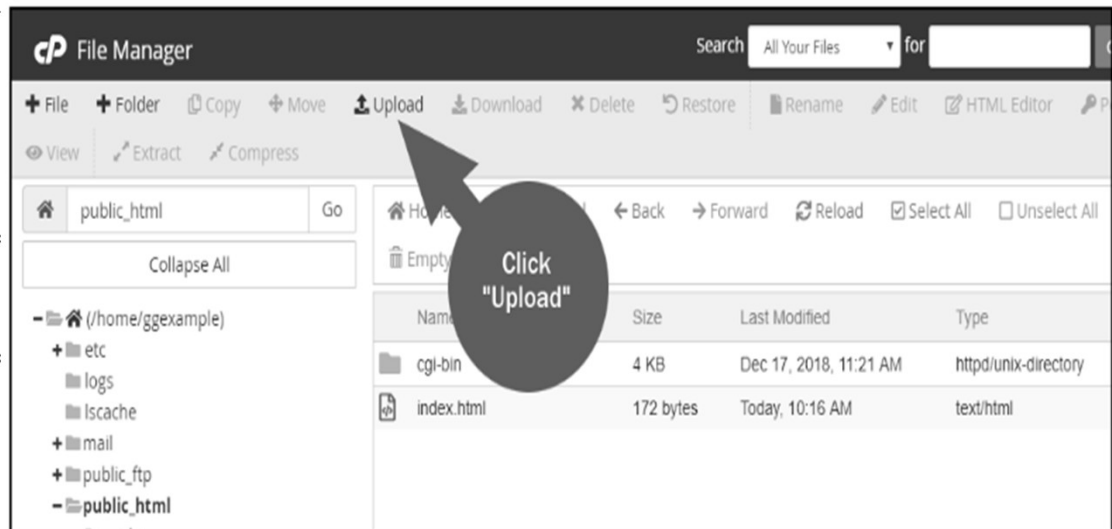


Accessing the cPanel -- 4

- **Step 4: Locating the Web Root (public_html)**
- On the left-hand sidebar of the File Manager, you will see a list of folders.
- Double-click the **public_html** folder.
- **Crucial Concept:** This is your "Web Root." Any file placed here is visible to the public. If you name a file `index.html` and put it here, it becomes your homepage.

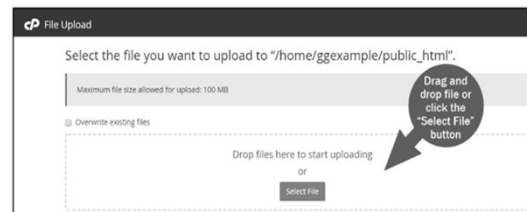


Click "Upload" in the top toolbar.



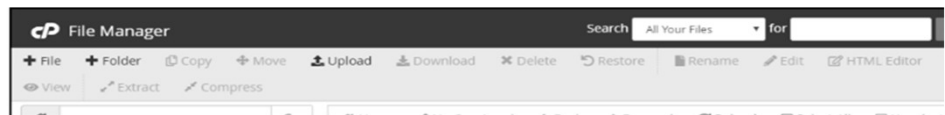
Accessing the cPanel -- 5

- **Step 5: Uploading Your Files**
- Click the **Upload** button in the top toolbar. A new browser tab will open.
- **Pro-Tip:** If you have many files (images, CSS, JS), **Zip** them on your computer first into one file (e.g., website.zip). cPanel handles one large zip file much faster than 100 small individual files.
- Drag and drop your file/zip into the box or click **Select File**.
- Wait for the progress bar to turn **Green (100%)**.



Accessing the cPanel -- 6

- **Step 6: Extracting and Organizing (If you uploaded a Zip)**
- Go back to the previous File Manager tab and click **Reload**.
- Right-click your website.zip and select **Extract**.
- Confirm the location (it should be /public_html).
- **Clean Up:** Once extracted, delete the .zip file to save space and stay organized.



Troubleshooting Checklist to Consider:

- **403 Forbidden Error:** Usually means you haven't uploaded an index.html or index.php file, or your folder permissions are wrong (directories should be 755).
- **Empty Site:** Ensure your files are *inside* public_html, not in a subfolder like public_html/my-website-folder/.

Uploading Files via FTP -- 1

- While the cPanel File Manager is excellent for quick edits and zipping files, professional developers also prefer FTP tools such as **FileZilla**.
- FileZilla is a dedicated FTP (File Transfer Protocol) client that allows you to manage files without keeping a web browser tab open, and it is much more stable for transferring large folders or thousands of images.

Uploading Files via FTP -- 2

- **1. Installation**
 - Download the **FileZilla Client** (ensure it is the "Client" version, not "Server") from the official website.
 - Run the installer. It works on Windows, macOS, and Linux – perfect for a diverse lab environment.
- **2. Gather Your Credentials**
 - To connect, you need four specific pieces of information from your hosting provider (usually found in your welcome email):
 - **Host:** Typically, **ftp.yourdomain.com** or your server's **IP Address**.
 - **Username:** Your cPanel username.
 - **Password:** Your cPanel password.
 - **Port:** Usually **21** (Standard FTP) or **22** (SFTP - Secure FTP).

Uploading Files via FTP -- 3

■ 3. The Quickconnect Method

- Open FileZilla.
- At the top, you will see the **Quickconnect** bar.
- Enter your **Host, Username, Password, and Port**.
- Click **Quickconnect**.
- If a window pops up regarding an "Unknown Certificate," check "Always trust this certificate" and click **OK**. This is common for servers using SSL for security.

■ 4. Understanding the Interface

- The screen is split into two main sections:
- **Left Side (Local Site):** These are the files currently on **your computer**.
- **Right Side (Remote Site):** These are the files currently on the **web server**.

Uploading Files via FTP -- 3

■ 5. Transferring Your Files

- On the **Right Side**, double-click the `public_html` folder to enter your web root.
- On the **Left Side**, navigate to the folder on your computer where your website code is saved.
- Highlight all the files you want to upload.
- **Drag and Drop** them from the left pane to the right pane.
- Watch the **Transfer Queue** at the bottom. Once it's empty and says "File transfer successful," your site is live!

cPanel or FileZilla (FTP)

Feature	cPanel File Manager	FileZilla (FTP)
Connection	HTTP (via Browser)	Dedicated FTP Protocol
Stability	May time out on large uploads	Can "Resume" interrupted transfers
Speed	Best for single .zip files	Best for many unzipped folders
Workflows	Good for quick "hotfixes"	Best for initial development and bulk uploads

Other FTP Tools and Troubleshooting Checklist

- Apart from FileZilla, other popular FTP tools include **WinSCP**, **Cyberduck**, and **Transmit**,
- **Troubleshooting Checklist: Common Connection Errors**
- If you cannot connect, check these three things immediately:
 1. **Firewall:** Is the institution or Windows firewall blocking Port 21?
 2. **Passive vs. Active Mode:** If the connection hangs at "Retrieving Directory Listing," switch the settings to **Passive Mode**.
 3. **Protocol:** Ensure they aren't trying to connect to an **SFTP** port (22) using the FTP protocol (21).

Note:

If you are replacing an existing file, check the “Overwrite existing files” box (if you don’t check the box you will get a pop-up asking if you want to overwrite the existing file). You can upload files up to 100 MB in size using File Manager. For larger files, use FTP.

Get more on web hosting from here:

<https://www.hostgator.in/blog/how-to-host-a-website-a-complete-guide-for-beginners/>